Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) REGULAR EXAMINATION
FOURTH SEMESTER

MG4DSEECT203 SINGLE BOARD COMPUTERS FOT IOT APPLICATIONS

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)

Part A

Multiple Choice Question

Answer all questions. Each question carries **1** mark

1.Arduino Uno is based on which microcontroller? [CO1]/[U]
a) ATmega16
b) ATmega328P
c) ESP8266
d) ARM Cortex

2.Raspberry Pi is a [CO2]/[U]
a) Microcontroller
b) Sensor
c) Single Board Computer
d) Actuator

3.OpenCV is mainly used for [C03]/[U]
a) Text editing
b) Image and video processing
c) Audio processing
d) Database handling

4.Which pin type is used to read analog values in Arduino Uno? [CO1]/[U]
a) Digital
b) PWM
c) Analog
d) Serial

5.Which component stores the operating system in Raspberry Pi? [CO2]/[U]
a) RAM
b) Processor
c) SD card
d) GPIO

6.Jetson Nano is mainly designed for [C03]/[U]
a) Web browsing
b) AI and Machine Learning
c) Office applications
d) Gaming

7.Which function sets a pin as INPUT or OUTPUT? [CO1]/[U]
a) digitalWrite()
b) pinMode()
c) analogRead()
d) delay()

8.Raspbian OS is based on [CO2]/[U]
a) Windows
b) Linux
c) macOS
d) DOS

9.Banana Pi belongs to which category? [C03]/[U]
a) Sensor
b) Actuator
c) Single Board Computer
d) Microcontroller

10.Which sensor is commonly used to measure distance? [CO1]/[U]
a) IR sensor
b) LDR
c) Ultrasonic sensor
d) Gas sensor

11.Which command is used to list files in Linux? [CO2]/[U]
a) cd
b) mkdir
c) ls
d) rm

12.BeagleBone is widely used for [C03]/[U]
a) Image editing
b) Real-time embedded applications
c) Word processing
d) Cloud storage

13.ESP8266 is mainly used for [CO1]/[U]
a) Bluetooth communication
b) Wi-Fi communication
c) GSM communication
d) Wired networking

14.Which Python library is used for GPIO control? [CO2]/[U]
a) NumPy
b) OpenCV
c) RPi.GPIO
d) Pandas

15.Home automation using Raspberry Pi is an example of [C03]/[U]
a) Desktop application
b) Manual system
c) IoT application
d) Offline system

16.Real-time sensor monitoring means [C03]/[U]
a) Storing data only
b) Manual reading
c) Continuous data observation
d) Periodic reporting

16X1=16

PART B
Short Answer Questions
Answer any 6 Questions. Each question carries 5 marks

17.Explain the pin configuration of Arduino Uno. [CO1]/[U]
18.Explain the hardware components of Raspberry Pi. [CO2]/[U]
19.Explain installation of OpenCV and camera interfacing. [CO3]/[U]
20.Explain the basic building blocks of IoT. [CO1]/[U]
21.Write short notes on basic Linux shell commands. [CO2]/[U]
22.Write a simple Python program to calculate area or volume. [CO2]/[U]
23.Write short notes on advanced SBCs: Tinkerboard and Banana Pi. [CO3]/[U]
24.Describe interfacing of LED and push button with Arduino. [CO1]/[U]

6X5=30

PART C
Essay Questions
Answer any 2 Questions. Each question carries 12 marks

25. Explain types of Arduino boards, Arduino IDE installation, and Embedded C programming with examples. [CO1]/[U]
26.Describe Raspberry Pi OS installation and explain GPIO control using Python with detailed explanation. [CO2]/[U]
27.Discuss NVIDIA Jetson Nano features and applications in AI and IoT. [CO3]/[U]
28.Explain IoT cloud platforms such as Blynk / Arduino Cloud / ThingSpeak and discuss real time IoT applications. [CO3]/[U]

2X12=24